Synthetic Organoid Generator

Parameter Reference Guide & Organoid Type Presets

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1. Overview

This tool generates realistic 3D synthetic fluorescence organoid images as OME-TIFF files, compatible with arivis Pro, FIJI, napari, and any OME-aware viewer. Two fluorescence channels are produced — DAPI nuclear stain and Actin/phalloidin membrane stain — along with two ground-truth label masks (cell bodies and nuclei individually).

The generator is controlled entirely through parameters, either via preset JSON files or by editing parameters.py directly. This document explains the biological meaning of each parameter group and how to tune them for different organoid types.

Output files per run

- organoid_<preset>_seed<N>_<timestamp>.ome.tif — 2-channel fluorescence image
- organoid_..._labels.ome.tif — cell body integer label mask (1 per cell)
- organoid_..._nucleus_labels.ome.tif — nucleus label mask (same IDs, nuclei only)

2. Parameter Groups Explained

Parameters are organised into seven groups. Each group controls a different physical aspect of the simulation. Understanding the biology behind each group is the key to creating convincing organoid-specific presets.

2.1 VoxelParams — Physical voxel dimensions

Sets the physical size of each voxel in micrometers. These values must match your microscope acquisition settings. All downstream size calculations (PSF blur, staining depth, cell radius) use these values to convert between voxels and micrometers.

Parameter	Group	Type	Default	Range	Description
voxel_size_xy	voxel	float (µm)	0.414	0.1–1.0	XY pixel size in µm. Match to your objective and camera pixel size.
voxel_size_z	voxel	float (µm)	1.0	0.3–5.0	Z step size in µm. Match to your z-stack acquisition step.

2.2 OrganoidShapeParams — Overall geometry

Controls the global organoid shape before any cells are placed. The organoid is modelled as an ellipsoid — a sphere optionally flattened along Z, mimicking the effect of gravity and substrate adhesion.

Parameter	Group	Type	Default	Range	Description
diameter_um	shape	float	150	50–300	Overall organoid size in µm. Larger = more cells, longer generation time.
diameter_std_um	shape	float	10	0–30	Run-to-run size variation. Set to 0 for reproducible exact size.
sphericity	shape	float	0.92	0.75–1.0	1.0 = perfect sphere. Lower = flattened along Z axis (oblate). Values below 0.85 give a clearly disc-like shape.

2.3 CellParams — Cell and nucleus morphology

This is the most important parameter group. It controls cell size, nucleus size relative to the cell, mechanical compression, nucleus shape, and architectural specialisations such as hollow lumens and necrotic cores. Most organoid-type differences are encoded here.

Cell size gradient

Real organoids have a clear size gradient: inner (core) cells are smaller due to mechanical confinement; outer (peripheral) cells are larger with more free surface. The ratio cell_radius_periph / cell_radius_core controls the steepness of this gradient.

Parameter	Group	Type	Default	Range	Description
cell_radius_core	cells	float (µm)	6.5	4–12	Radius of core cells. Below 4 µm, nuclei become unresolvable at typical voxel sizes.
cell_radius_periph	cells	float (µm)	10.0	6–16	Radius of peripheral cells. Ratio to core controls steepness of the radial gradient.
cell_radius_std	cells	float (µm)	0.6	0–2	Cell-to-cell size variation (lognormal sigma). Higher = more heterogeneous population.
core_fraction	cells	float	0.55	0.3–0.75	Fraction of organoid radius defined as the core zone. Also sets the ramp start for apical_elongation.

Nucleus-to-cell (NC) ratio

The NC ratio (linear: nucleus radius / cell radius) determines how much of the cell cross-section is occupied by the nucleus. In confocal images this is the most visually striking property. Carcinoma cells typically have high NC ratios (0.70–0.82); normal epithelium is lower (0.55–0.65). Volume ratio = NC ratio cubed.

Parameter	Group	Type	Default	Range	Description
nc_ratio_core	cells	float	0.72	0.50–0.82	NC ratio in core. Higher = nuclei fill more of each cell.
nc_ratio_periph	cells	float	0.70	0.48–0.80	NC ratio at periphery. Usually slightly lower than core.

Nucleus elongation

Real nuclei are not perfect spheres. Columnar epithelial cells have elongated nuclei with a short/long axis ratio below 1.0. This is independent of the cell body elongation controlled by apical_elongation.

Parameter	Group	Type	Default	Range	Description
elongation_core	cells	float	0.88	0.75–1.0	Nucleus short/long axis ratio in core. 1.0 = sphere. Floor is 0.75.
elongation_periph	cells	float	0.78	0.65–1.0	Nucleus elongation at periphery. Lower = more elongated (columnar epithelium).

Pressure model — mechanical compression

Each cell is modelled as a pressurised sphere with an inflated radius. Overlapping inflated spheres form contact patches. The larger the overlap, the larger and flatter the contact patch — directly simulating mechanical pressure in tightly packed tissue.

- pressure = 0.0: cells just touch at a point — maximum roundness, visible intercellular gaps
- pressure = 0.15: moderate contact patches — typical for a packed organoid
- pressure = 0.35: large flat contact zones — heavily compressed, necrotic-core appearance

Parameter	Group	Type	Default	Range	Description
pressure_core	cells	float	0.18	0.0–0.40	Compression of core cells. Higher = more compressed, flatter contact faces. Tumour spheroids: 0.22–0.35.
pressure_periph	cells	float	0.06	0.0–0.20	Compression of peripheral cells. Usually 3–4× lower than core.
radial_compression	cells	float	0.15	0.0–0.30	Oblate flattening of core cells along the radial axis. 0 = sphere, 0.25 = noticeably oblate.

Nucleus shape, irregularity, and eccentricity

Real nuclei range from nearly round (hepatocytes, stem cells) to highly irregular kidney/bean shapes (carcinoma cells under stress). The eccentricity parameters model nuclear displacement toward one side of the cell — especially in peripheral cells where the nucleus is pushed outward. The direction of eccentricity blends smoothly from random in core cells to radially outward in peripheral cells using a t^2 weight function.

Parameter	Group	Type	Default	Range	Description
nucleus_irregularity	cells	float	0.25	0.0–0.40	Surface deformation of nucleus. 0 = perfect ellipsoid, 0.35 = strong kidney/bean shape (high-grade carcinoma).
nucleus_ecc_core	cells	float	0.08	0.0–0.20	Nuclear offset in core cells as fraction of r_cell. Direction is random (no polarity).
nucleus_ecc_periph	cells	float	0.22	0.0–0.35	Nuclear offset in peripheral cells. Direction ramps from random to radially outward via t^2 blend.

Necrotic core

Large organoids (>150 µm) develop a hypoxic/necrotic core where inner cells lose membrane integrity, nuclei condense, and actin signal collapses. Three nuclear phenotypes are assigned deterministically per cell: pyknotic (65%, condensed/bright), ghost cells (20%, nearly full-size/very dim — nucleus largely dissolved), and karyorrhectic (15%, partially fragmented). Actin signal is set to zero for all necrotic-zone cells.

Parameter	Group	Type	Default	Range	Description
necrotic_core	cells	bool	false	true/false	Enable the necrotic zone. When true, inner cells receive three-phenotype DAPI treatment and zero actin.
necrotic_fraction	cells	float	0.25	0.05–0.50	Inner radius fraction assigned to the necrotic zone. 0.25 = innermost 25% of organoid radius.
necrotic_dapi_boost	cells	float	1.6	1.0–3.0	Brightness multiplier for pyknotic and karyorrhectic nuclei. Ghost cell dimness is fixed independently.

Hollow lumen (cyst mode)

When lumen_fraction > 0, a true geometric exclusion zone is created during cell packing — no cells are seeded or relaxed into the inner fraction of the organoid radius. This leaves a genuine void that renders as background fluorescence, correctly representing cyst and hollow-shell organoid architectures. A centrifugal repulsion force prevents cells from drifting back into the lumen during relaxation.

Parameter	Group	Type	Default	Range	Description
lumen_fraction	cells	float	0.0	0.0–0.80	Hollow lumen radius as fraction of organoid radius. 0 = solid spheroid. 0.78 = thin polarised shell (cyst mode). Do not combine with necrotic_core.

Apical-basal cell elongation and surface flattening

Real epithelial cells are taller radially than tangentially (columnar) and the outermost cells flatten against the ECM/air interface (squamous-like). These parameters control the shape of each cell's power-diagram territory in the radial direction, independently of nucleus elongation. Apical elongation ramps smoothly from zero at core_fraction to full value at the organoid surface using a t^2 weight. Surface flattening acts only in the outermost 15% of the organoid radius. When both parameters are 0.0, cell territories are isotropic — identical to the previous behaviour.

Parameter	Group	Type	Default	Range	Description
apical_elongation	cells	float	0.0	0.0–0.60	Radial stretch of peripheral cell territories. 0 = isotropic. 0.25 = mild columnar. 0.50 = clearly columnar crypt. Ramps from core_fraction to surface via t^2 weight.
surface_flattening	cells	float	0.0	0.0–0.60	Radial compression of the outermost cells (outermost 15% of radius). 0 = none. 0.30 = visible squamous outer layer. Mimics cells pressed against ECM or air interface.

2.4 PackingParams — Force-directed cell placement

Controls how cells settle into their packed configuration. The generator uses a physics simulation: cells repel each other when overlapping and are pushed back inside the organoid boundary. More iterations gives tighter, more uniform packing. Most users do not need to change these.

Parameter	Group	Type	Default	Range	Description
n_iterations	packing	int	120	60–300	Force-directed relaxation steps. More = tighter packing, slower generation.
repulsion_strength	packing	float	0.45	0.2–0.8	Repulsion force between overlapping cells. Higher = more uniform spacing.
damping	packing	float	0.55	0.3–0.8	Velocity damping per relaxation step. Lower = faster settling, higher = more stable.
target_overlap	packing	float	0.02	0.0–0.10	Target fractional overlap between adjacent cells after relaxation. Controls tightness of final packing.
boundary_strength	packing	float	1.2	0.5–2.0	Force keeping cells inside the organoid boundary. Also scales lumen repulsion (×1.5) when lumen_fraction > 0.

2.5 TextureParams — Signal texture and membrane appearance

Controls fine-grained visual texture: chromatin patterns inside nuclei, membrane thickness, zone-based brightness gradients, cell-to-cell brightness variation, and the curvature of cell-cell boundaries.

Parameter	Group	Type	Default	Range	Description
nucleus_texture_scale	texture	float	0.20	0.05–0.50	Spatial frequency scale of the chromatin texture noise field. Lower = coarser texture, higher = finer granularity.

Parameter	Group	Type	Default	Range	Description
nucleus_texture_amplitude	texture	float	0.35	0.0–0.60	Strength of chromatin texture inside nuclei. 0 = uniform fill, 0.6 = strongly granular.
nucleus_texture_octaves	texture	int	3	1–5	Number of fractal octaves in the chromatin noise. More octaves = more complex multi-scale texture.
membrane_thickness_vox	texture	float	2.0	1.0–4.0	Actin membrane thickness in voxels. Higher = thicker, more prominent membranes.
actin_cytoplasm_frac	texture	float	0.08	0.0–0.25	Fraction of membrane brightness in the cytoplasm. Higher = more filled-in cells.
membrane_bend_amplitude	texture	float	0.20	0.0–0.35	Low-frequency noise displacement applied before power-diagram competition. 0 = flat Voronoi faces. 0.20 = biologically natural curves. Critical for avoiding straight-edge artefacts.
dapi_intensity_sigma	texture	float	0.18	0.0–0.40	Cell-to-cell DAPI brightness variation (lognormal sigma). Higher = more heterogeneous staining.
actin_intensity_sigma	texture	float	0.28	0.0–0.45	Cell-to-cell actin brightness variation.
dapi_brightness_core	texture	float	0.70	0.3–1.0	Relative DAPI brightness scaling for core cells. Values below 1.0 produce a dimmer interior consistent with staining gradient effects.
dapi_brightness_periph	texture	float	1.00	0.5–1.0	Relative DAPI brightness scaling for peripheral cells. Typically 1.0 (full brightness at organoid surface).
actin_brightness_core	texture	float	0.55	0.2–1.0	Relative actin brightness scaling for core cells. Lower values reflect reduced membrane signal in compressed inner cells.
actin_brightness_periph	texture	float	1.00	0.5–1.0	Relative actin brightness scaling for peripheral cells. Typically 1.0.

2.6 OpticsParams — Microscope optics simulation

Simulates the physical optics of the microscope. These parameters have the largest effect on how realistic the image looks. The PSF blur is the single most important factor — it should match the actual microscope used for real data collection.

Parameter	Group	Type	Default	Range	Description
psf_sigma_xy_um	optics	float (µm)	0.25	0.15–0.50	Lateral PSF Gaussian sigma. Match to objective NA: 0.20 = high NA confocal, 0.40 = widefield.
psf_sigma_z_um	optics	float (µm)	1.20	0.5–3.0	Axial PSF sigma. Always larger than XY: 0.65 = lightsheet, 1.2 = confocal, 2.0 = widefield.
z_attenuation_coeff	optics	float	0.005	0.001–0.015	Overall intensity falloff with depth. Match to sample thickness and clearing status.

Parameter	Group	Type	Default	Range	Description
haze_sigma_um	optics	float (µm)	10.0	5.0–30.0	Spatial scale of out-of-focus haze blur kernel in µm. Larger = broader diffuse glow from adjacent planes.
haze_amplitude	optics	float	0.10	0.0–0.25	Amplitude of out-of-focus haze. Higher = more diffuse background.
shot_noise_scale	optics	float	0.05	0.01–0.12	Poisson photon noise scale. Higher = grainier, lower SNR.
read_noise_std	optics	float	0.012	0.003–0.030	Gaussian detector readout noise standard deviation. Adds a constant noise floor independent of signal level.
background_level	optics	float	0.04	0.01–0.10	Constant background offset. Higher = brighter background as in uncleared samples.
crosstalk_fraction	optics	float	0.04	0.0–0.15	Bidirectional channel bleed-through between DAPI and actin channels.
staining_depth_um	optics	float (µm)	35.0	5–9999	Half-penetration depth of dye from organoid surface. Profile: 0.85 × exp(−d/depth) + 0.15. The 0.15 plateau reflects incomplete dye washout in uncleared samples. 9999 = uniform staining.
scatter_increase_rate	optics	float	0.005	0.0–0.012	PSF broadening per µm depth: σ_xy(z) = psf_sigma_xy × (1 + rate × z_µm). 0 = cleared/lightsheet, 0.003 = confocal uncleared, 0.008 = widefield thick sample.
cleared	optics	bool	false	true/false	Tissue-clearing flag. Disables staining gradient and depth-dependent PSF broadening. Use the apply_clearing() convenience method.

2.7 OutputParams — Output volume dimensions

Controls the voxel dimensions of the output volume. In normal use these are set automatically by auto_size_volume(), which computes the required volume from the organoid diameter with a 1.25× margin. Override manually only if you need a specific volume size.

Parameter	Group	Type	Default	Range	Description
vol_x	output	int	256	64–1024	Volume width in voxels (X dimension). Auto-set by auto_size_volume().
vol_y	output	int	256	64–1024	Volume height in voxels (Y dimension). Auto-set by auto_size_volume(). Always equal to vol_x.
vol_z	output	int	80	32–512	Volume depth in voxels (Z dimension). Auto-set by auto_size_volume().
bit_depth	output	int	16	8 or 16	Output bit depth. 16-bit is strongly recommended for downstream quantitative analysis.

3. How to Choose Parameters for a New Organoid Type

Start with the closest existing preset and adjust the parameters that differ biologically. The sections below give practical guidance for the most important decisions.

3.1 Cell size and gradient

- Large gradient (core much smaller than periphery): intestinal crypts, columnar epithelia, organoids under high mechanical confinement. Set cell_radius_core = 5–6, cell_radius_periph = 9–11 (ratio ~1.7).
- Shallow gradient (core and periphery similar): tumour spheroids, liver organoids, early-stage undifferentiated organoids. Set cell_radius_core = 7–8, cell_radius_periph = 8–10 (ratio ~1.2).

3.2 NC ratio

- High NC ratio (0.72–0.82): cancer cells (breast, pancreatic, colon carcinoma). Large irregular nuclei almost filling the cell.
- Medium NC ratio (0.62–0.72): most organoid types. Normal proliferating epithelium.
- Low NC ratio (0.50–0.62): hepatocytes, mature differentiated cells. Large cytoplasm, relatively small nucleus.

3.3 Pressure

- High pressure (core 0.22–0.35): dense tumour spheroids, hypoxic cores, cells under mechanical stress.
- Low pressure (core 0.08–0.14): brain organoids, loosely packed structures, early-stage organoids with ECM.

3.4 Nucleus irregularity

- Low (0.10–0.18): normal epithelium, hepatocytes, stem-cell organoids. Round to mildly oval nuclei.
- High (0.28–0.40): cancer cells, cells under stress. Kidney/bean/multilobular shapes. Hallmark of malignant transformation.

3.5 Hollow lumen vs necrotic core

- lumen_fraction > 0: use for cyst-forming organoids where cells form a shell around a fluid-filled void. No cells are present in the lumen — it renders as background. Do not combine with necrotic_core = true.
- necrotic_core = true: use for large solid tumour organoids (>150 µm) where the core is necrotic but cells are physically present. The three-phenotype DAPI distribution (pyknotic bright / ghost dim / karyorrhectic intermediate) is applied automatically.

3.6 Apical-basal polarity

- Set apical_elongation > 0 for genuinely polarised epithelium — intestinal crypt, kidney tubular, cyst-forming lines. Typical values: 0.20–0.40.
- Set surface_flattening > 0 when the outermost cell layer should look squamous — normal mammary epithelium, organoids with a well-defined ECM interface. Typical values: 0.15–0.35.
- Leave both at 0.0 for cancer spheroids and non-polarised cell lines — these cells have lost epithelial organisation and show no consistent radial shape gradient.

3.7 Optics settings by modality

- Confocal (e.g. Nikon AX, ZEISS LSM 980): psf_sigma_xy = 0.20–0.25, psf_sigma_z = 0.90–1.40, scatter_increase_rate = 0.003
- Lightsheet (e.g. ZEISS Lightsheet 7): psf_sigma_xy = 0.18–0.22, psf_sigma_z = 0.50–0.90, scatter_increase_rate = 0.0
- Widefield / lightfield: psf_sigma_xy = 0.30–0.50, psf_sigma_z = 1.50–3.00, scatter_increase_rate = 0.005–0.008
- Spinning disk confocal: psf_sigma_xy = 0.20–0.28, psf_sigma_z = 0.80–1.20, scatter_increase_rate = 0.002–0.004

4. Available Presets

All presets are JSON files in the presets/ folder. Run any preset from the command line or load it in the GUI. Parameters not specified in the JSON file inherit their defaults from parameters.py.

4.1 General-purpose presets

Preset	Diam (µm)	~Cells	Cell size core→periph (µm)	Pressure core/periph	NC ratio	Nuc irreg	Character
tiny_test	55	~23	6.5 → 10.0	0.18 / 0.06	0.72 / 0.70	0.25	Fast testing only, ~5 s. Not intended for biological use.
tumor_spheroid	120	~200	6.5 → 8.0	0.25 / 0.07	0.73 / 0.70	0.22	Dense HCT116/MCF7-like spheroid. High core pressure.
pdac_organoid	160	~350	7.0 → 10.0	0.16 / 0.05	0.74 / 0.70	0.30	Pancreatic ductal adenocarcinoma. Heterogeneous, pleomorphic.
intestinal_crypt	130	~280	5.5 → 7.0	0.22 / 0.08	0.75 / 0.72	0.18	Columnar epithelium. Elongated nuclei, large core-to-periph gradient.
breast_cancer	140	~300	7.0 → 9.5	0.28 / 0.09	0.78 / 0.75	0.35	Pleomorphic, high NC ratio, disorganised packing.
brain_organoid	180	~250	9.0 → 13.0	0.10 / 0.03	0.65 / 0.62	0.32	Large loose cells. Very irregular nuclei, low pressure.
prostate_cancer	120	~220	6.0 → 8.5	0.20 / 0.07	0.70 / 0.68	0.22	Glandular architecture, moderate pleomorphism.
hepatic_organoid	160	~220	10.0 → 13.0	0.12 / 0.04	0.62 / 0.60	0.14	Large hepatocyte-like cells. Round nuclei, low NC ratio.
kidney_organoid	150	~290	6.5 → 9.0	0.18 / 0.06	0.68 / 0.65	0.20	Tubular epithelium, moderately elongated nuclei.

4.2 Calibrated Ong et al. 2025 presets

Five presets calibrated against measured per-cell feature data from Ong et al. (2025) (Nature Methods, DOI: 10.1038/s41592-025-02685-4). Imaging conditions: Nikon Ti2-E AX confocal, 25× Plan Apo Lambda S silicone 1.05 NA objective, 0.345 µm pixel, 1.0 µm z-step, NucBlue + Alexa Fluor 647 Phalloidin, Fluoromount G mounting. Calibration targets included nuclear density, crystal distance, Ripley ratio, and cell volume distributions matched to the published per-cell data for each condition.

Preset	Diam (µm)	~Cells	Cell size core→periph (µm)	Pressure core/periph	NC ratio	Nuc irreg	Character
pdac_isotonic	170	~640	7.0 → 9.5	0.18 / 0.06	0.72 / 0.68	0.26	PDAC spheroid, isotonic control. Cell volume ~2300 µm³, elongation 1.40. heterochromatin_fraction = 0.30 (heterogeneous open chromatin).
pdac_hypertonic	165	~610	7.0 → 9.5	0.22 / 0.08	0.72 / 0.68	0.30	PDAC spheroid, osmotic stress. Slight nuclear volume decrease.

Preset	Diam (µm)	~Cells	Cell size core→periph (µm)	Pressure core/periph	NC ratio	Nuc irreg	Character
							heterochromatin_fraction = 0.05 (compacted uniform chromatin).
hmecyst_control	195	~260	12.0 → 14.5	0.12 / 0.05	0.65 / 0.62	0.14	HMECyst normal epithelial (A01). Nuclear density 94,869/mm³. Crystal distance 22.7 µm. Solid spheroid, Ripley ratio ~1.7.
hmecyst_cyst	190	~40–60	10.5 → 13.0	0.08 / 0.20	0.65 / 0.63	0.12	HMECyst cyst-forming (A07). lumen_fraction = 0.78. Crystal distance 17.3 µm. Ripley ratio 2.721 (crystal-like shell).
pdac_large_clustering	200	~1000	6.5 → 10.0	0.28 / 0.09	0.74 / 0.70	0.34	Large primary PDAC. necrotic_core = true, necrotic_fraction = 0.28. Continuous radial gradient; intensity and size features dominate cluster separation.

5. Basis for Preset Parameter Values

Parameters fall into three categories by evidential basis: Category A (directly grounded in published quantitative measurements), Category B (qualitatively motivated by published biology, representing reasoned estimates), and Category C (purely algorithmic with no biological interpretation).

5.1 Parameters with strong literature support (Category A)

NC ratio

The NC ratio is a quantitative diagnostic criterion used in clinical cytopathology. Published imaging flow cytometry measurements consistently find NC ratios of 0.60–0.70 for cancer cell lines. Non-malignant breast epithelium (MCF-10A) measures ~0.53. Our presets use cancer values of 0.70–0.78 and normal epithelium values of 0.62–0.68.

Cell size

Cell diameters are well established in histology. Hepatocytes: 20–30 µm diameter (cell_radius 10–13 µm). Typical carcinoma cells: 15–25 µm diameter. Intestinal columnar epithelial cells: 8–15 µm diameter, 20–40 µm height. The core-to-periphery size gradient is a conserved feature of all organoids.

Nucleus shape irregularity

Nuclear pleomorphism is a formal grading criterion in tumour pathology (WHO Classification of Tumours): smooth round nuclei (Grade 1), mild irregularity (Grade 2), marked pleomorphism (Grade 3). Our nucleus_irregularity parameter (0.10–0.40) maps directly to this scale.

Necrotic core

The necrotic core is a well-documented feature of spheroids above approximately 150 µm diameter. The inner core exhibits pyknotic nuclei and absent actin signal. The three-phenotype nuclear population (pyknotic, ghost, karyorrhectic) reflects established necrosis pathology: pyknosis and karyorrhexis are recognised histological stages preceding karyolysis (ghost cell phenotype).

Staining gradient

The staining attenuation model uses $0.85 \times \exp(-d/depth) + 0.15$, where d is distance from the organoid surface. The 0.15 residual plateau reflects incomplete washout of phalloidin and DAPI in uncleared organoids — real confocal images of thick uncleared samples consistently show a dim but non-zero interior signal.

5.2 Reasoned estimates (Category B)

Pressure values

Tumour spheroids develop internal hydrostatic compressive stress of approximately 1–10 kPa. Brain tissue Young's modulus is 0.1–1 kPa. These relative orderings motivate the pressure parameter choices, but the absolute values should be calibrated against morphometric measurements from real images.

Apical elongation and surface flattening

These parameters have no published quantitative equivalent — they are specific to the power-diagram-based cell territory model. Values in calibrated presets were tuned visually by comparing generated center-plane slices against confocal images of the respective organoid types.

Optics parameters

PSF sigma values are instrument-specific. Values in each preset represent typical ranges for the named modality class. Your specific microscope will have different values — measure them from a bead calibration sample.

5.3 Algorithmic parameters (Category C)

Packing parameters (n_iterations, repulsion_strength, damping, target_overlap, boundary_strength) control numerical convergence of the sphere packing algorithm and have no biological interpretation. Texture noise frequencies and octave counts are aesthetic choices producing visually realistic variation and do not correspond to specific biological length scales.

5.4 Recommended validation approach

The most rigorous way to validate and refine parameters is to:

- Generate synthetic organoids with each preset
- Run them through a segmentation pipeline (e.g. Cellpose 3D, StarDist-3D)
- Extract morphometric statistics: NC ratio, nuclear volume, elongation, cell-cell distance, Ripley ratio
- Compare against published measurements for that organoid type
- Iteratively adjust preset parameters until distributions match

6. Creating a Custom Preset

Create a new JSON file in the presets/ folder. Include only parameters that differ from defaults — everything else is inherited from parameters.py. Then add the preset name to AVAILABLE_PRESETS in generate_organoid.py.

Example — custom lung adenocarcinoma organoid:

```
// File: presets/lung_adeno.json {   "_description": "Lung adenocarcinoma organoid",
"shape": { "diameter_um": 130.0, "sphericity": 0.91 },   "cells": {
"cell_radius_core": 6.0, "cell_radius_periph": 9.0,     "nc_ratio_core": 0.74,
"nc_ratio_periph": 0.71,     "pressure_core": 0.20, "pressure_periph": 0.07,
"nucleus_irregularity": 0.28,     "apical_elongation": 0.20, "surface_flattening": 0.25
} }
```

Then register the preset and run:

```
python generate_organoid.py --preset lung_adeno --seed 42 python generate_organoid.py --
preset lung_adeno --batch 20 --output-dir ./lung_dataset
```

A _calibration block is recommended to document the biological measurements and imaging conditions used to derive parameter values. See existing calibrated presets for examples.

7. Command-Line Reference

Flag	Type	Default	Range	Description	Example
--preset	str	None	see §4	Load a parameter preset JSON.	--preset pdac_isotonic
--seed	int	42	any int	Random seed. Same seed + preset = identical output.	--seed 7
--diameter	float	from preset	50–400 µm	Override organoid diameter without editing the preset file.	--diameter 180
--ncells	int	—	1–5000	Target approximate cell count (sets diameter automatically).	--ncells 500
--voxel-xy	float	0.414	0.1–1.0	XY pixel size in µm. Match to your acquisition settings.	--voxel-xy 0.345
--voxel-z	float	1.0	0.3–5.0	Z step size in µm.	--voxel-z 1.0
--output	path	auto	any	Explicit output path. If omitted, auto-named with timestamp.	--output my_organoid.ome.tif
--output-dir	path	output/	any	Directory for auto-named files. Created if it does not exist.	--output-dir D:/data
--batch	int	1	1–1000	Generate N organoids with sequential seeds.	--batch 20
--no-labels	flag	off	—	Skip saving label masks (faster, smaller output).	--no-labels
--gpu	flag	off	—	GPU acceleration via CuPy for the optics stage.	--gpu